



Introduction to Bioengineering Theory (BBL1020; 3-0-0)

Lecture-20



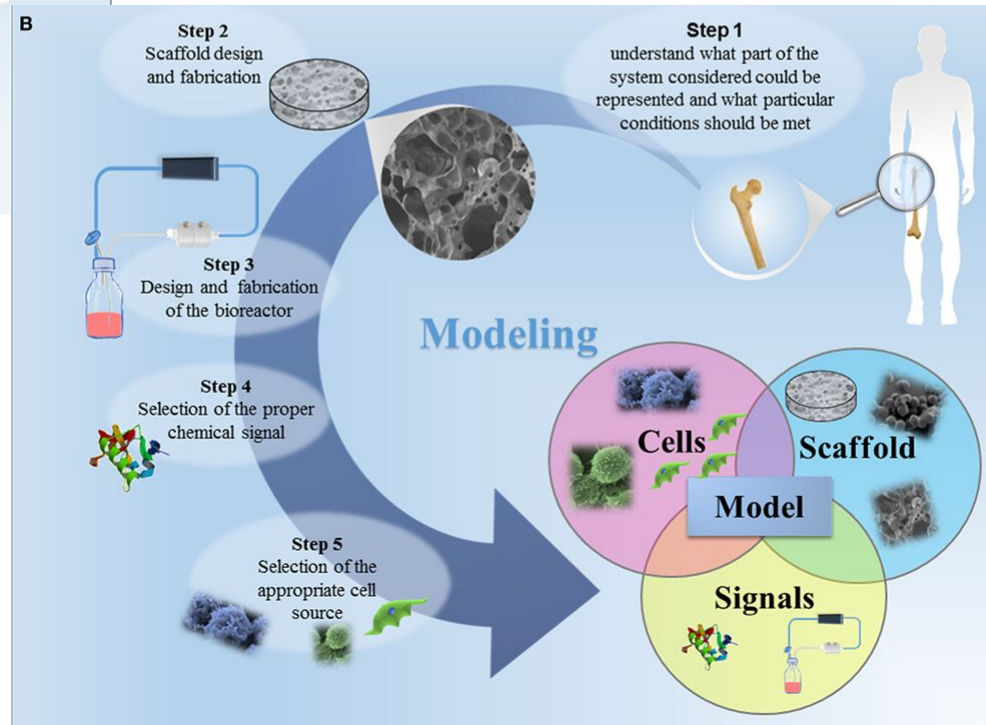
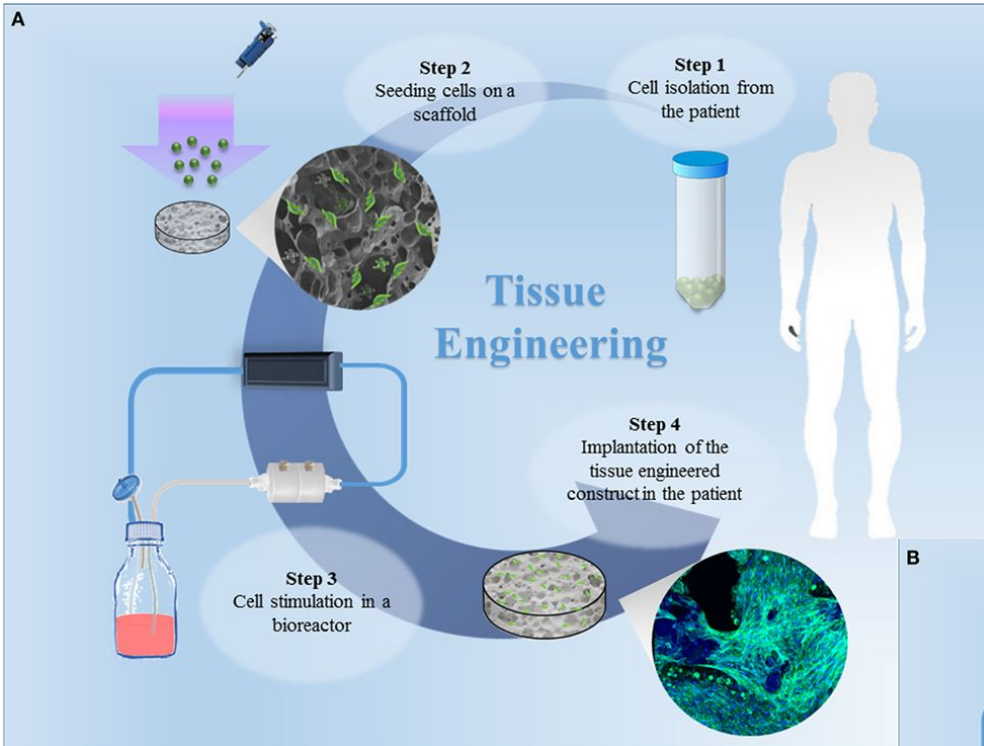
Tissue Engineering & Biomaterials

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Tissue engineering

- Tissue engineering is an interdisciplinary field that applies the principles of engineering and the life sciences toward the development of biological substitutes that restore, maintain, or improve tissue function. Three general strategies have been adopted for the creation of new tissue:
 - **Isolated cells or cell substitutes:** This approach avoids the complications of surgery, allows replacement of only those cells that supply the needed function, and permits manipulation of cells before infusion. Its potential limitations include failure of the infused cells to maintain their function in the recipient, and immunological rejection.
 - **Tissue-inducing substances:** The success of this approach depends on the purification and large-scale production of appropriate signal molecules, such as growth factors, and, in many cases, the development of methods to deliver these molecules to their targets.
 - **Cells placed on or within matrices:** In closed systems, the cells are isolated from the body by a membrane that allows permeation of nutrients and wastes but prevents large entities such as antibodies or immune cells from destroying the transplant.
- In clinical practice this term is closely related to the applications that repair or replace portions/whole tissues (i.e., bone, cartilage, blood vessels, skin, bladder etc.)

Basic protocol



Traditional tissue engineering vs. in situ regeneration

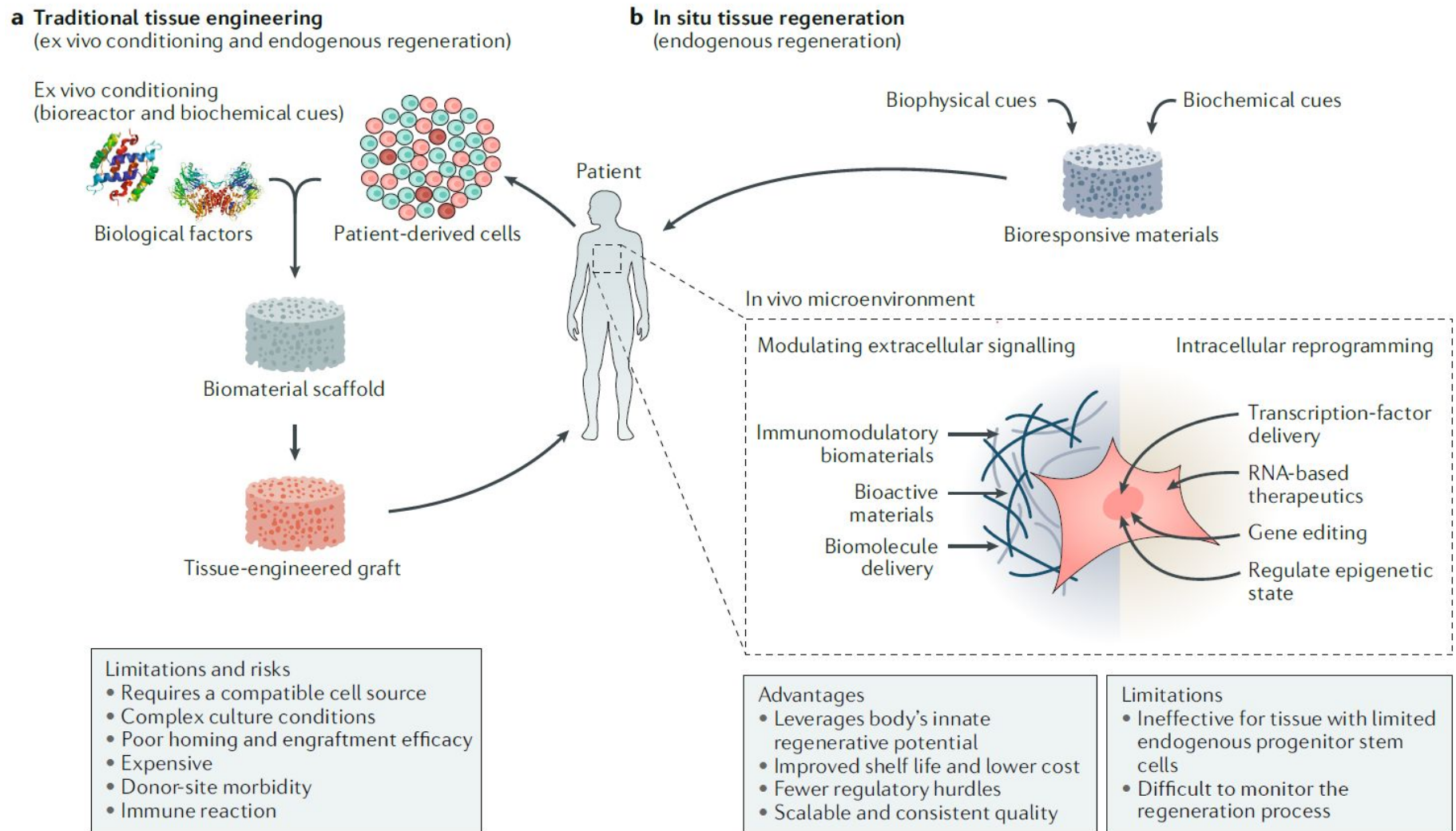
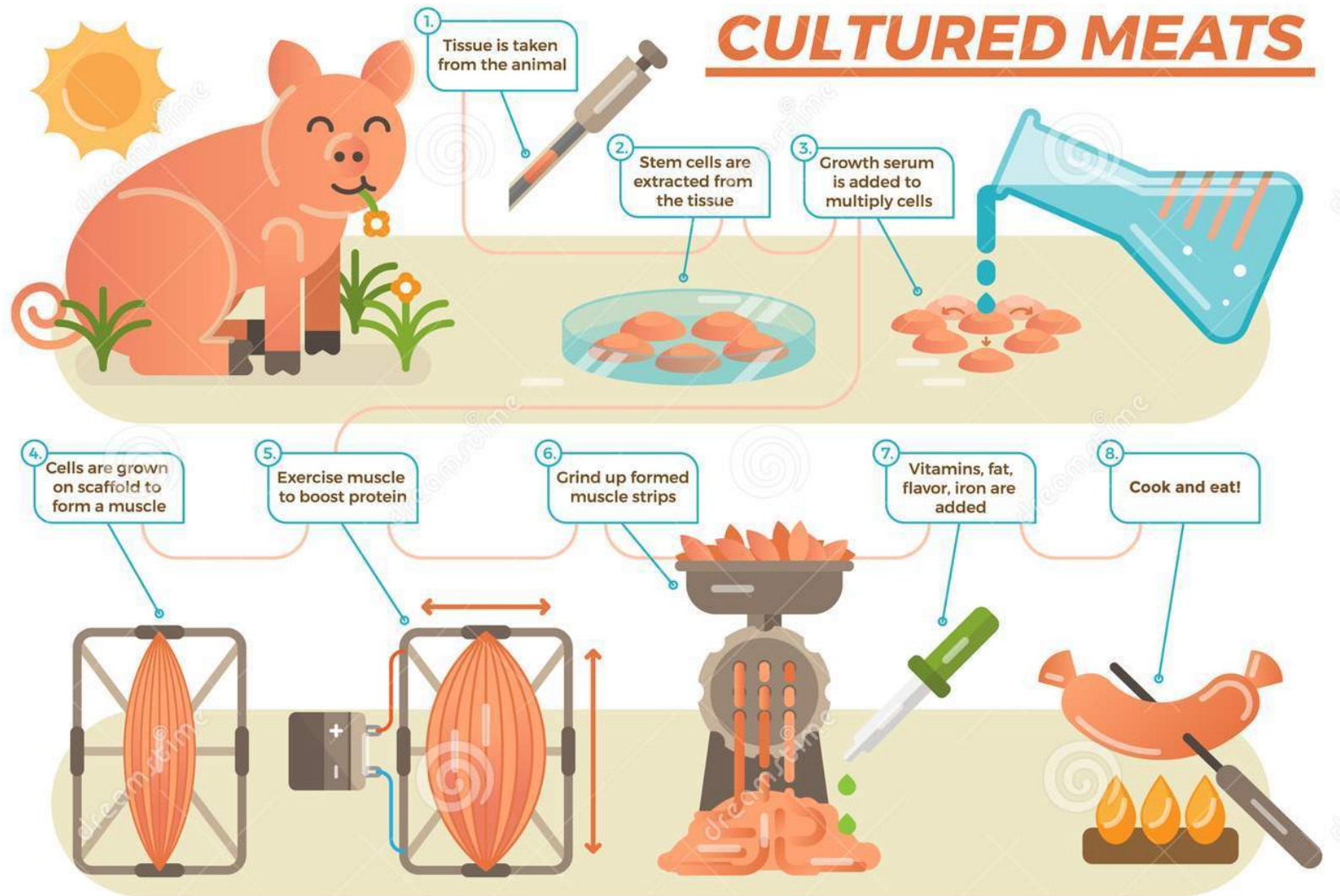


Fig. 1 | **In situ tissue engineering.** **a** | Traditional tissue-engineering approaches require the pre-seeding of engineered scaffolds and ex vivo conditioning before implantation into the body. **b** | In situ tissue regeneration uses bioresponsive materials that harness the innate regenerative ability of the body. These materials are loaded with biochemical and biophysical cues to recruit endogenous cells for tissue healing.

Clean meat using tissue engineering



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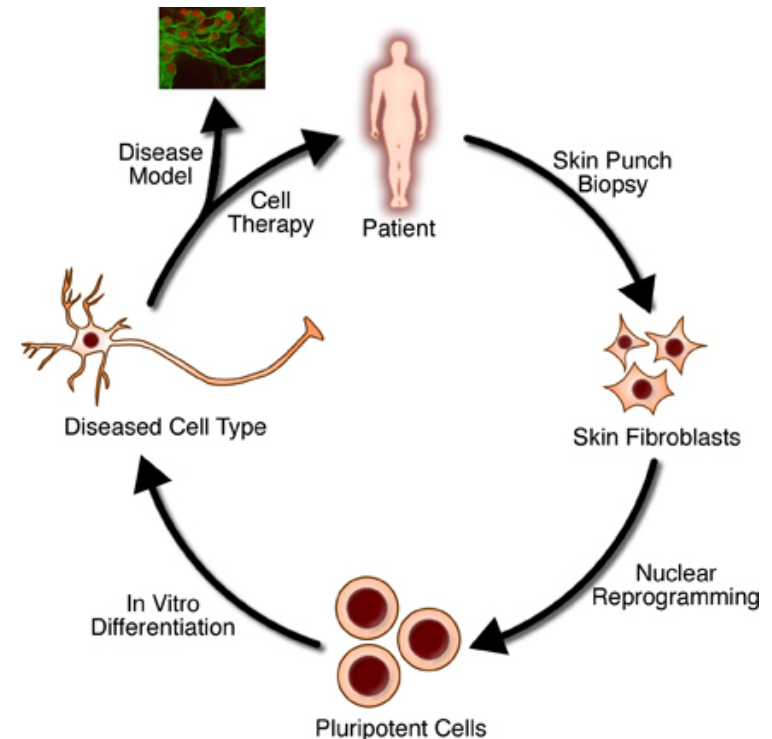
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Regenerative medicine

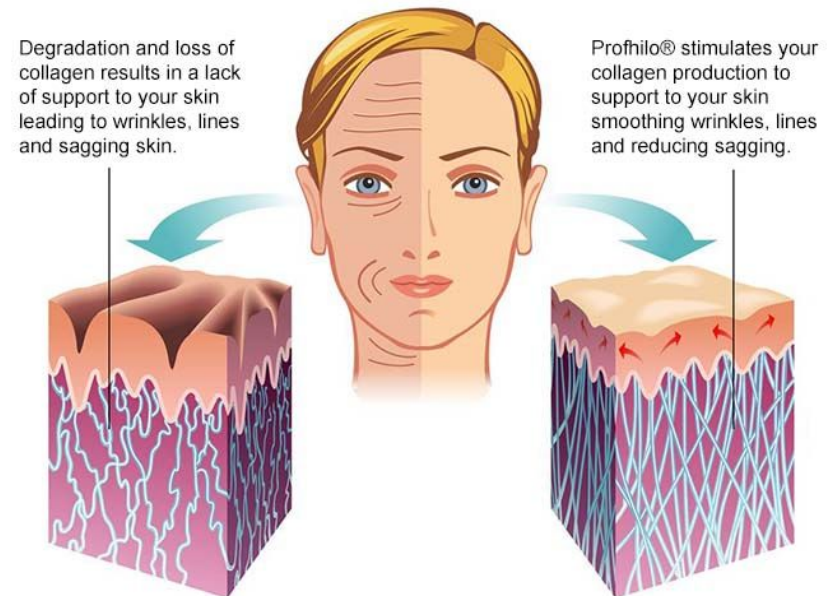
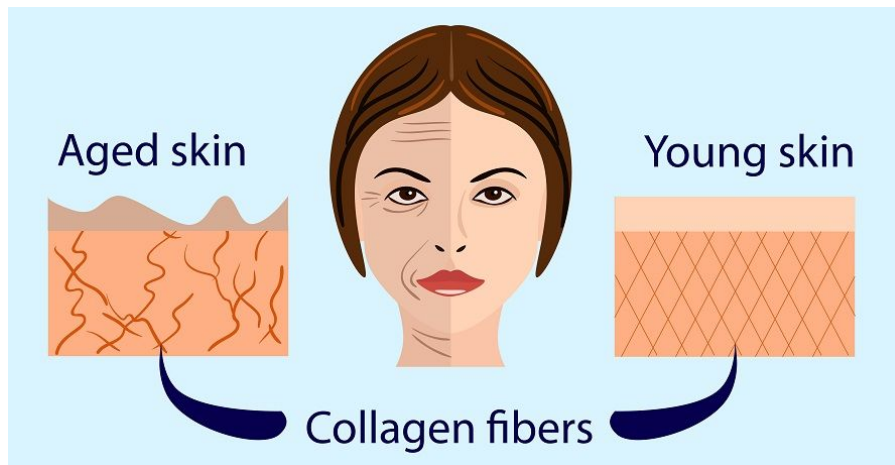
- Regenerative medicine deals with the "process of replacing, engineering or regenerating human or animal cells, tissues or organs to restore or establish normal function".
- This field holds the promise of engineering damaged tissues and organs by stimulating the body's own repair mechanisms to functionally heal previously irreparable tissues or organs.
- When the cell source for a regenerated organ is derived from the patient's own tissue or cells, the challenge of organ transplant rejection via immunological mismatch is circumvented. This approach could alleviate the problem of the shortage of organs available for donation.
- Some of the biomedical approaches within the field of regenerative medicine may involve the use of stem cells. Examples include the injection of stem cells or progenitor cells obtained through directed differentiation (cell therapies); the induction of regeneration by biologically active molecules administered alone or as a secretion by infused cells (immunomodulation therapy); and transplantation of in vitro grown organs and tissues (tissue engineering).



Regenerative medicine based FDA approved products

Table 1. Regenerative medicine FDA-approved products

Category	Name	Biological agent	Approved use
Biologics	laViv	Autologous fibroblasts	Improving nasolabial fold appearance
	Carticel	Autologous chondrocytes	Cartilage defects from acute or repetitive trauma
	Apligraf, GINTUIT	Allogeneic cultured keratinocytes and fibroblasts in bovine collagen	Topical mucogingival conditions, leg and diabetic foot ulcers
	Cord blood	Hematopoietic stem and progenitor cells	Hematopoietic and immunological reconstitution after myeloablative treatment
Cell-based medical devices	Dermagraft	Allogenic fibroblasts	Diabetic foot ulcer
	Celution	Cell extraction	Transfer of autologous adipose stem cells
Biopharmaceuticals	GEM 125	PDGF-BB, tricalcium phosphate	Periodontal defects
	Regranex	PDGF-BB	Lower extremity diabetic ulcers
	Infuse, Infuse bone graft, Inductos	BMP-2	Tibia fracture and nonunion, and lower spine fusion
	Osteogenic protein-1	BMP-7	Tibia nonunion



Biomaterials – a general overview

History of Biomaterials

- **1860's** – Lister develops aseptic surgical technique
- **Early 1900's**: Bone plates used to fix fractures.
- **1930's**: Introduction of cobalt chromium alloys.
- **1938**: First total hip prosthesis
- **1940's**: Polymers in medicine: PMMA bone repair, cellulose for dialysis, nylon sutures.
- **1952**: Mechanical heart valve
- **1953**: Dacron [polymer fiber] vascular grafts
- **1958**: Cemented [PMMA] joint replacement
- **1960**: first commercial heart valve
- **1970's**: PEO protein resistant thin film coating
- **1976**: FDA amendment governing testing & production of biomaterials / devices
- **1976**: Artificial heart

Definition of a Biomaterial

- A material intended to interface with biological systems to **evaluate, treat, augment, or replace** any tissue, organ or function of the body

(The Williams Dictionary of Biomaterials, 1999)

[OR]

- A biomaterial is a substance that has been engineered to take a form which, **alone or as part of a complex system**, is used to **direct, by control of interactions with components of living systems**, the course of any therapeutic or diagnostic procedure, in human or veterinary medicine”

(Williams, D. F. 2009. On the nature of biomaterials. *Biomaterials*, 30(30), p. 5897-5909).

What is the need for a Biomaterial?

- Millions of patients suffer end stage organ and tissue failure annually.
- \$400 billion annually for treatment.
- 8 million surgical procedures.
- The biomaterials market is poised to reach USD 130.57 Billion by 2020, growing at a CAGR of 16% during the forecast period of 2015 to 2020. (<http://www.marketsandmarkets.com/Market-Reports/biomaterials-393.html>)
- Treatment options include transplantation, reconstruction, mechanical devices.

Available options and limitations

■ Transplantation

- Critical donor shortage
 - 3000 livers annually for 30000 people in need
- Disease Transmission
 - HIV
 - Hepatitis
 - Other transmittable diseases.

■ Surgical Reconstruction

- Not always possible
- Complications

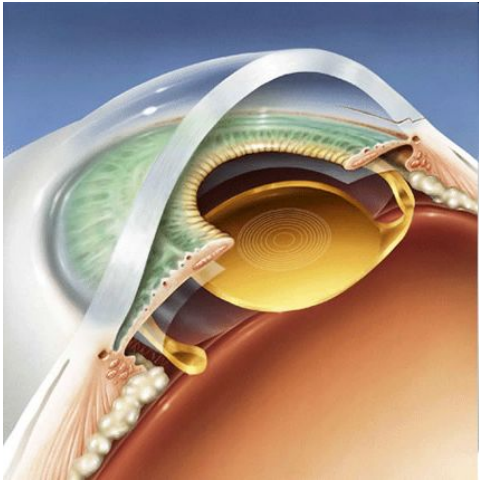
■ Mechanical Devices

- Engineering approach – engineer new tissues, systems
- Limited by
 - Complexity of the human body
 - Multiple functions
 - Living components versus non living components
 - Materials?
 - Tissue based
 - Polymers
 - Metals
 - Ceramics

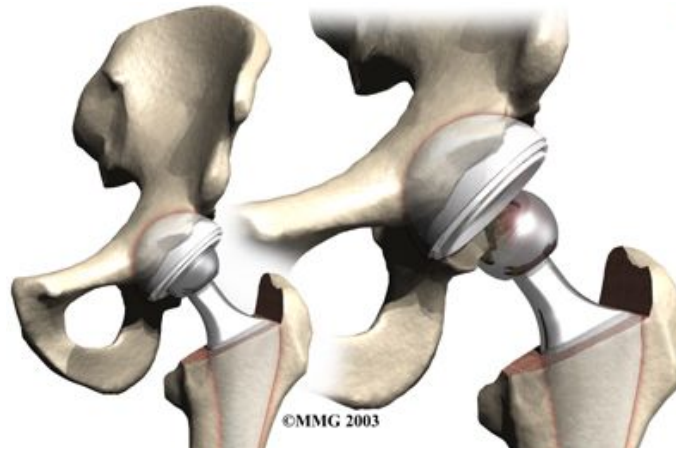
Commonly used biomaterials

<u>Material</u>	<u>Applications</u>
■ Silicone rubber	Catheters, tubing
■ Dacron	Vascular grafts
■ Cellulose	Dialysis membranes
■ PMMA	Intraocular lenses, bone cement
■ Polyurethanes	Catheters, pacemaker leads
■ Hydrogels	Ophthalmological devices, Drug Delivery
■ Stainless steel	Orthopedic devices, stents
■ Titanium	Orthopedic and dental devices
■ Alumina	Orthopedic and dental devices
■ Hydroxyapatite	Orthopedic and dental devices
■ Collagen (reprocessed)	Ophthalmologic applications, wound dressings

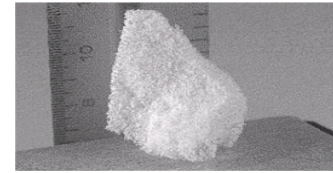
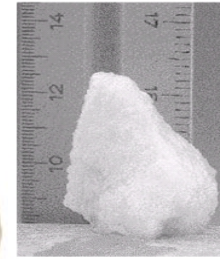
Examples



Intraocular lens
(PMMA, acrylic, silicone)



Artificial hip joints
(Stainless steel)



In the shape of a nose [left] is seeded with cells called chondrocytes that replace the polymer with cartilage over time [right] to make a suitable implants.



Biological valve
(human or porcine)

Mechanical valve



Heart valves

- **Biocompatibility:** The ability of a material to perform with an appropriate host response in a specific application.
- **Host response:** the reaction of a living system to the presence of a material

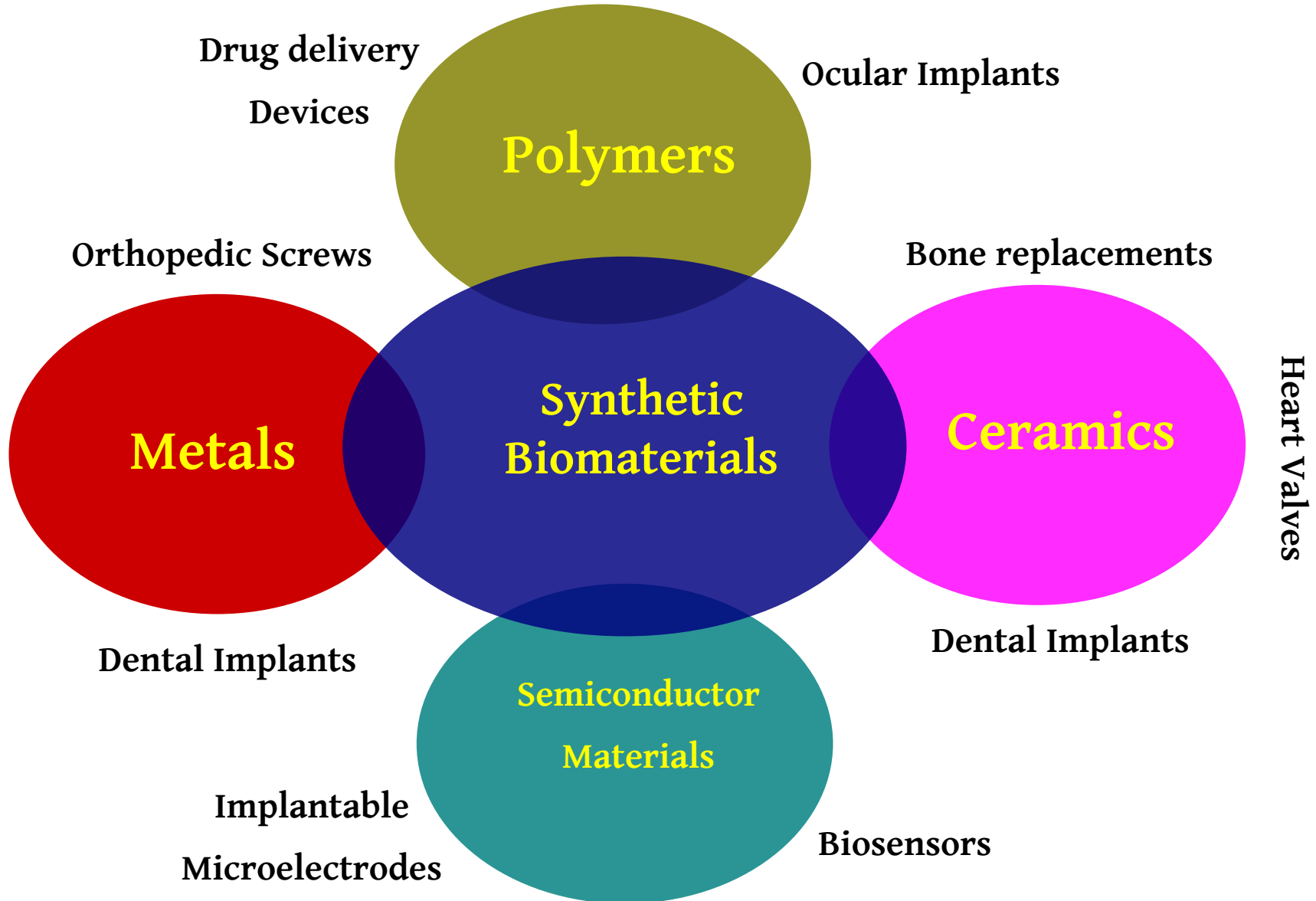
General criteria for material selection

- Mechanical and chemicals properties
- No undersirable biological effects
carcinogenic, toxic, allergenic or immunogenic
- Possible to process, fabricate and sterilize with a good reproducibility
- Acceptable cost/benefit ratio

Material properties

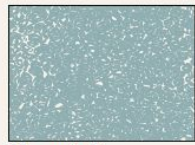
- Compressive strength
- Tensile strength
- Bending strength
- E-Modulus
- Coefficient of thermal expansion
- Coefficient of thermal conductivity
- Creep
- Surface tension
- Hardness and density
- Hydrophobic/phillic
- Water sorption/solubility
- Surface friction
- Bonding properties

Classification of biomaterials

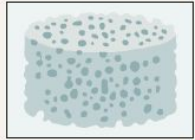


Approaches to engineer biomaterials

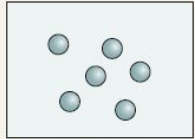
a Scaffold types



Monolithic



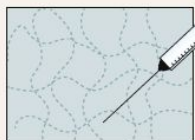
Microporous



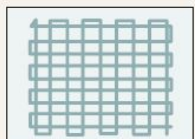
Nanoparticles



Fibrous



Hydrogel network

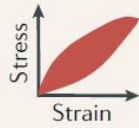


3D printed

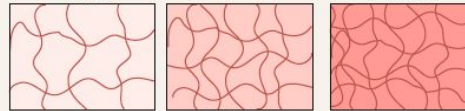
b Biophysical characteristics

Bulk properties

Viscoelasticity

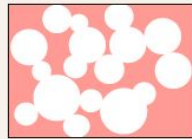


Soft → Mechanical stiffness → Stiff



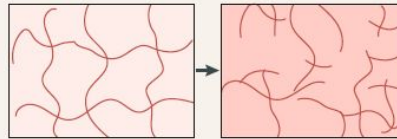
Provide mechanical signalling to cells

Architecture



Facilitate endogenous-cell infiltration and scaffold remodelling

Degradation



Surface properties

Guidance cues

Alignment

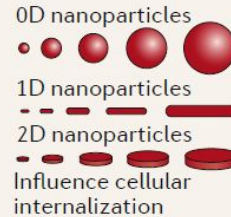


Pattern



Provide contact guidance, cell alignment and migration

Shape and size



Influence cellular internalization

Surface roughness



Surface charge



Wetting characteristics



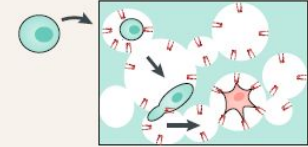
Influence cell adhesion, proliferation, migration and scaffold remodelling

c Biochemical characteristics

Chemical structure

- Metals
- Ceramics
- Polymers
- Composites

Cell-adhesion ligands



Facilitate endogenous cell homing

Cytokines, antigens and growth factors



Control and direct differentiation of endogenous cells

Mineral (ions)



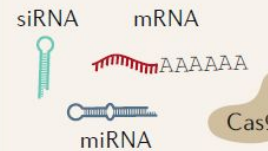
Immunomodulatory effect, regulate cell fate

Anti-inflammatory agents



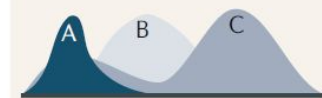
Suppress inflammatory response owing to biomaterials implantation

Reprogramming factors



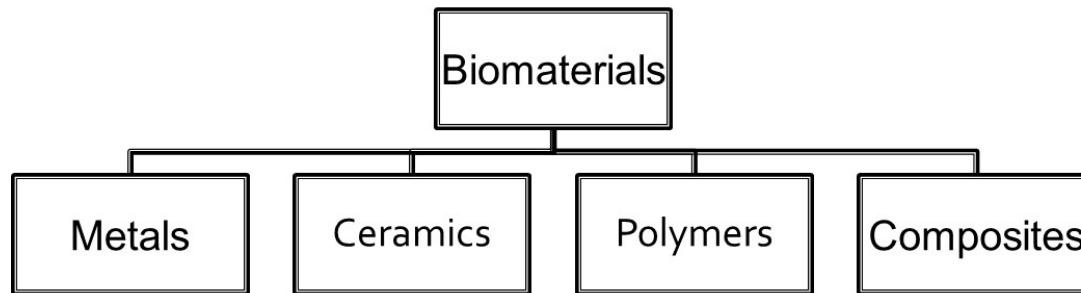
Induce differentiation or transdifferentiation of endogenous cells

Spatio-temporal representation



Control and facilitate endogenous cell migration and differentiation

Types of Biomaterials



Materials	Advantages	Disadvantages	Applications
Metals Titanium alloys Cobalt -chromium alloys Stainless steel Gold Platinum alloy	High impact strength High resistance to wear Ductile High conductivity	Low biocompatibility May corrode Low mechanical strength, high cost	Joint replacement, Bone plates and screws, Dental root implant Neuromuscular stimulation
Ceramics Alumina Zirconia Carbon Calcium phosphates	Good biocompatibility Inert, corrosion resistance Strong in compression Biodegradable	Brittle, Not resilient, Degradation not controllable	Dental implant, Femoral head of hip replacement, Coating of dental and orthopedic implants Drug delivery system

Types of Biomaterials (contd.)

Materials	Advantages	Disadvantages	Applications
Polymers Nylon, Polyethylene, Silicone, Teflon, Dacron, Acrylates, PGA, PLA	Low density Easy to fabricate	Low mechanical strength Deforms with time, may degrade	Sutures, vascular graft, hip socket, intraocular lenses Drug delivery system (DDS)
Composites Carbon-carbon Ceramic-polymer	Strong, less stiff than metals, Strong in compression	Difficult to make, Weak in tension	Joint implants Heart valve

Biomaterial testing

- Physical and mechanical
- Biological

In vitro assessment

in vivo assessment

Functional assessment

Clinical assessment

